EXPLORATORY DATA ANALYSIS

E-commerce Customer & Transaction Dataset

Task 5 — Data Analyst Internship, DataX Labs

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Tools used:	Python (Pandas, Matplotlib, Seaborn)
Dataset size:	1,000 records x 12 columns
Deliverables:	Jupyter Notebook + PDF Report

1. Objective

Extract patterns, trends, relationships, and anomalies from an e-commerce customer and transaction dataset using visual and statistical exploration techniques, and summarize actionable findings for the business.

2. Dataset Overview

The dataset contains 1,000 customer transaction records with 12 columns spanning demographics (Age, Gender, City), purchase behaviour (ProductCategory, Quantity, Price, TotalSpend, DaysSinceLastPurchase), and engagement metrics (SpendingScore, Rating).

Column	Type	Description
CustomerID	Text	Unique customer identifier
Age	Numeric	Customer age in years
Gender	Categorical	Male / Female
City	Categorical	Customer city (8 cities)
AnnualIncome	Numeric	Estimated annual income (Rs.)
SpendingScore	Numeric	Engagement/spending score (1-100)
ProductCategory	Categorical	Category of item purchased
Quantity	Numeric	Units purchased in the order
Price	Numeric	Unit price of the item (Rs.)
Rating	Numeric	Customer rating given (1-5)
DaysSinceLastPurchase	Numeric	Recency in days
TotalSpend	Numeric	Price x Quantity (Rs.)

Data quality note: AnnualIncome and Rating each had ~2% missing values, imputed with the column median. A small number of intentional outliers exist in AnnualIncome and Price to reflect realistic high-value customers/orders.

3. Univariate Analysis

3.1 Distributions (Histograms)

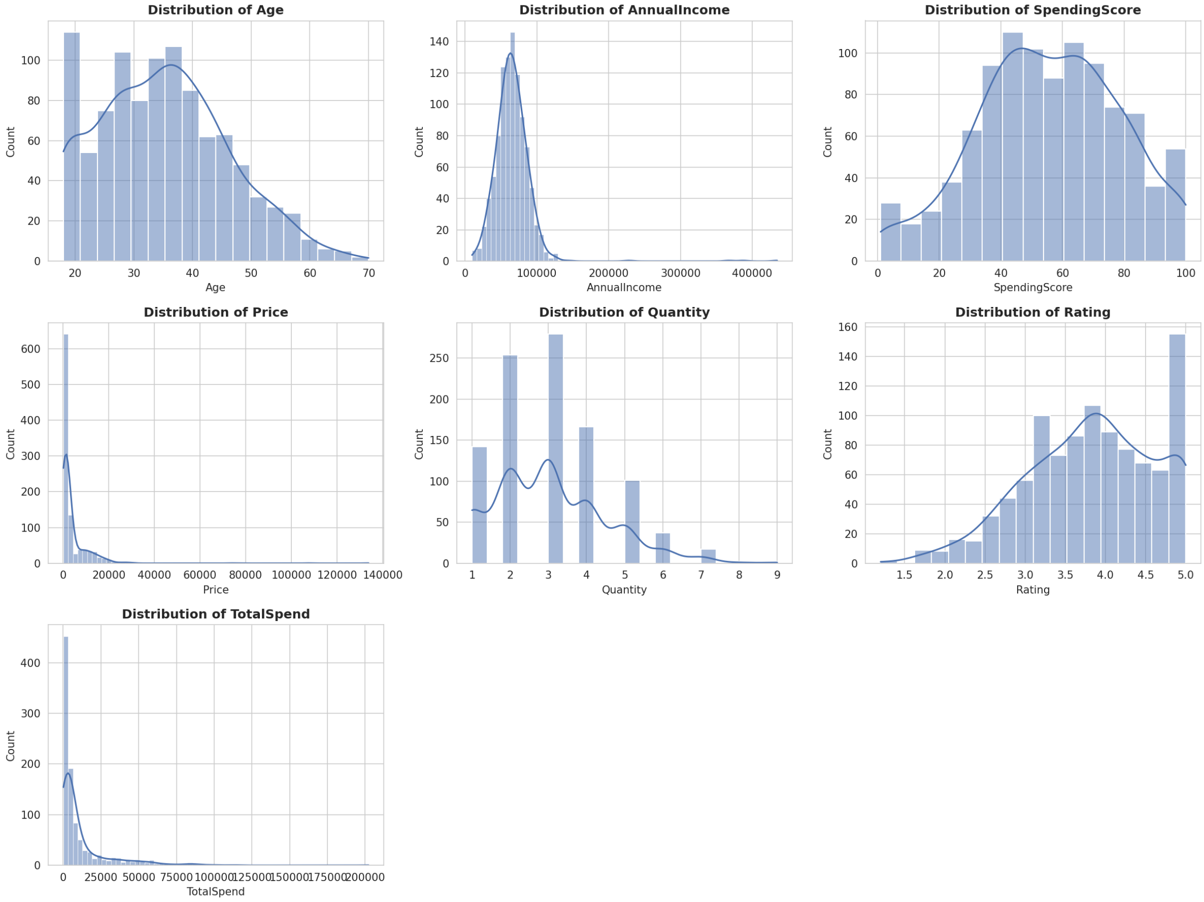

Figure 1: Distribution of key numeric variables.

Age, SpendingScore and Rating are approximately symmetric/normal. AnnualIncome, Price and TotalSpend are right-skewed, with a small set of high-value outliers pulling the tail out. Quantity is discrete and right-skewed, with most orders between 1-4 units.

3.2 Outlier Detection (Boxplots)

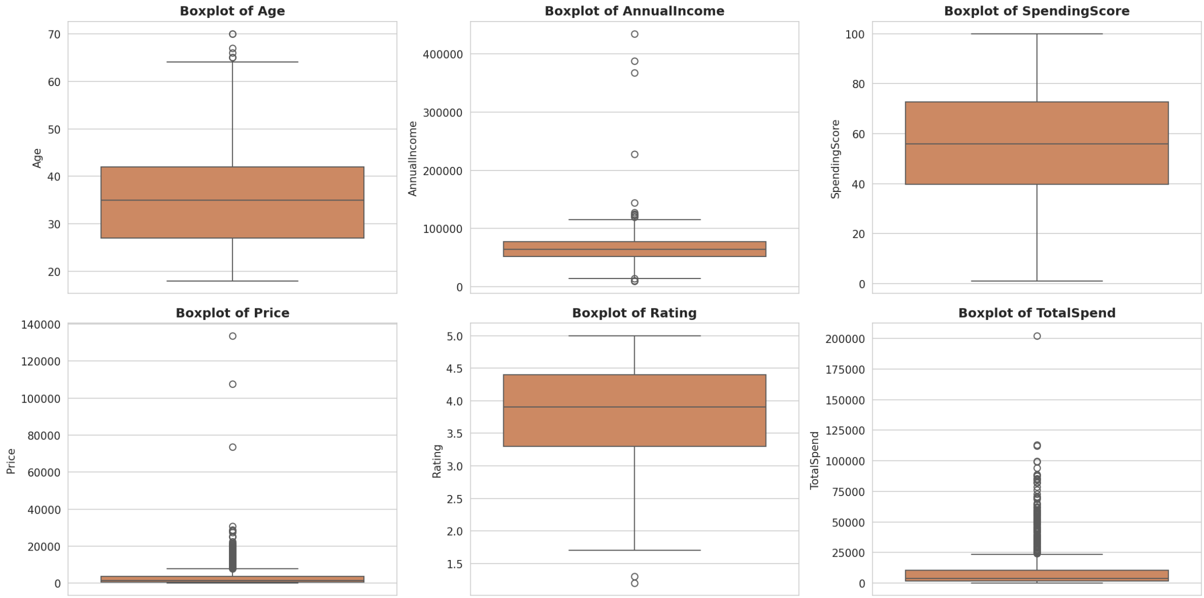

Figure 2: Boxplots highlighting spread and outliers.

AnnualIncome and Price (and therefore TotalSpend) show clear outliers above the upper whisker. These likely represent genuine high-value customers/premium orders rather than data errors, so they were flagged for review rather than removed.

3.3 Categorical Breakdown

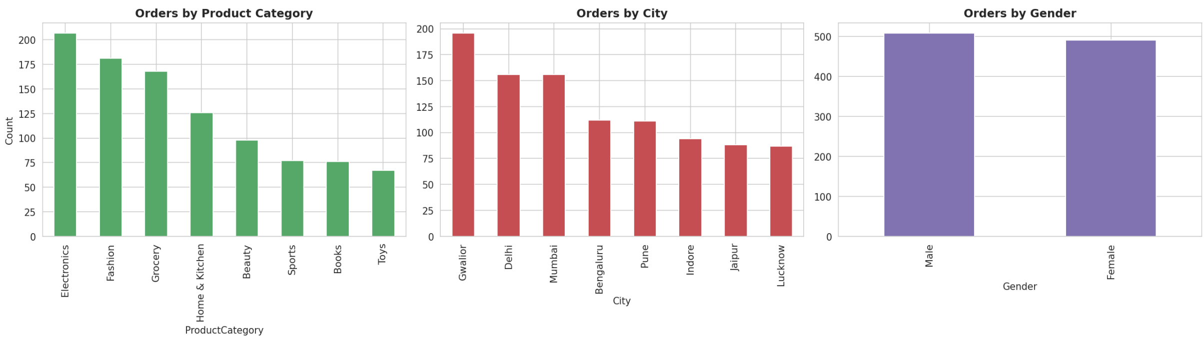

Figure 3: Order counts by Product Category, City and Gender.

Electronics and Fashion are the top two categories by order volume. Gwalior leads by city share, followed by Delhi and Mumbai. Gender split is close to balanced (52% Male / 48% Female).

4. Bivariate Analysis

Figure 4: Age vs SpendingScore, Income vs SpendingScore, Price vs Quantity.

Annual Income vs Spending Score shows a mild negative trend: higher earners do not necessarily register a higher spending score. Age vs Spending Score shows no strong pattern across either gender. Price vs Quantity shows no clear relationship, implying purchase quantity is driven more by need than by affordability.

4.1 Spend by Product Category

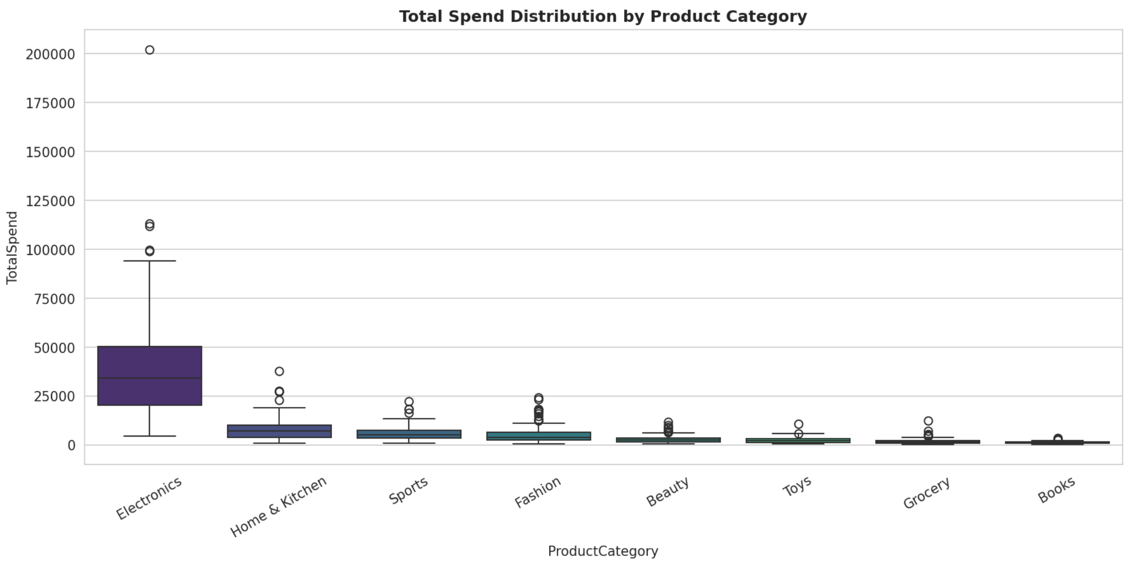

Figure 5: Total spend distribution across product categories.

Electronics has both the highest median spend and widest spread, consistent with higher unit prices and the top outliers. Books and Grocery cluster at the low end with tight, consistent spend values.

5. Multivariate Analysis

5.1 Pairplot

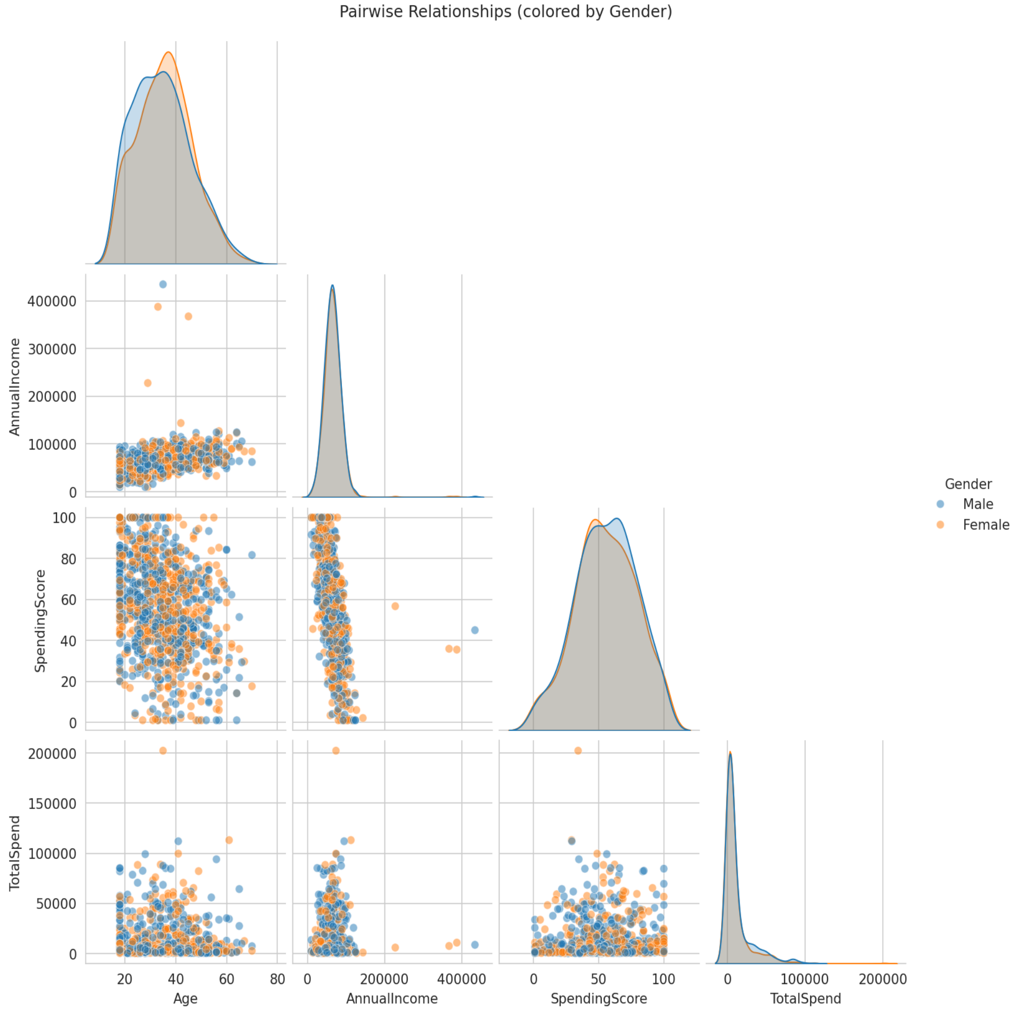

Figure 6: Pairwise relationships between Age, Income, Spending Score and Total Spend, by gender.

5.2 Correlation Heatmap

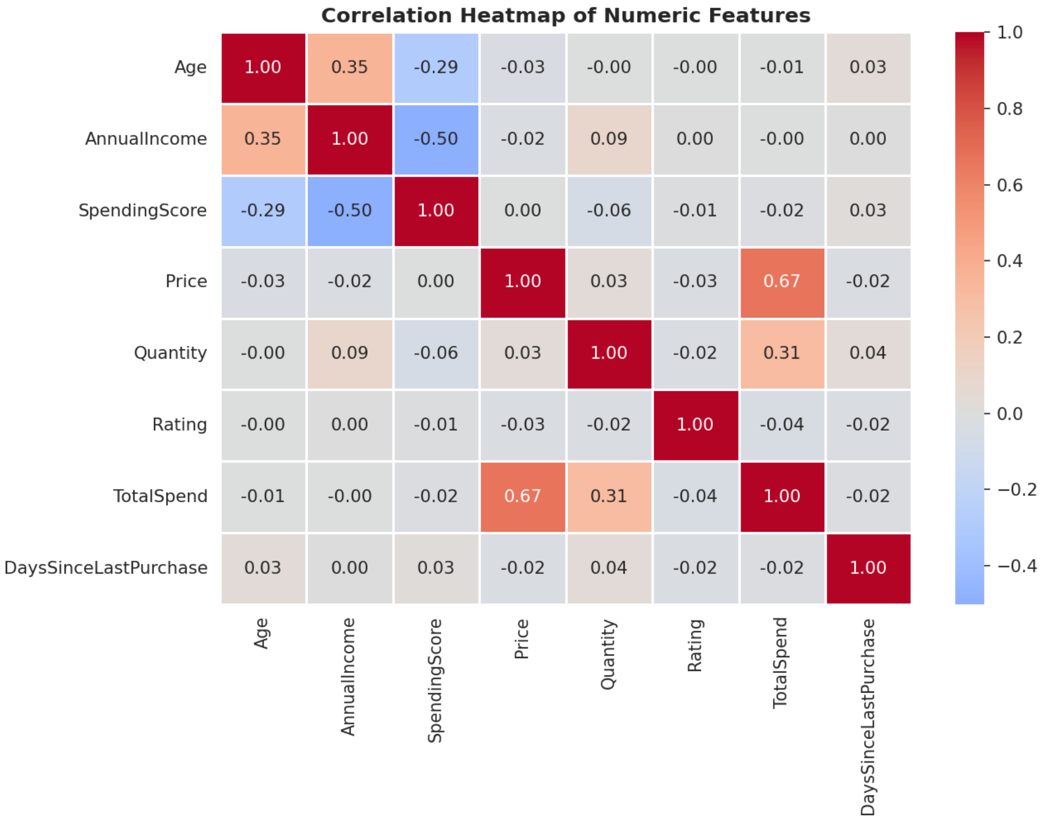

Figure 7: Correlation matrix of numeric features.

TotalSpend correlates strongly with Price (r ~ 0.67) and moderately with Quantity (r ~ 0.31), as expected since TotalSpend = Price x Quantity. AnnualIncome shows a moderate negative correlation with SpendingScore (r ~ -0.50). Age, Rating and DaysSinceLastPurchase are weakly correlated (|r| < 0.05) with other features, indicating low multicollinearity risk if used together in a model.

6. Summary of Findings

- Data quality is good: only ~2% missing values in AnnualIncome and Rating, all other fields are complete; missing values were imputed with the median.
- AnnualIncome, Price and TotalSpend are right-skewed with genuine high-value outliers worth a business review (possible VIP customers / premium orders).
- Electronics and Fashion dominate order volume; Electronics drives the highest revenue per order and the widest spend variance.
- AnnualIncome and SpendingScore show a moderate negative relationship (r ~ -0.50) - spending behaviour is not purely income-driven.
- Most numeric features show low correlation with each other, meaning low multicollinearity risk for downstream modelling.
- Gwalior contributes the largest share of orders in this sample, followed by Delhi and Mumbai.

7. Recommendation / Next Step

Since spending behaviour is not purely income-driven, customer segmentation should combine SpendingScore, ProductCategory affinity and DaysSinceLastPurchase (recency) rather than income alone. A natural next step is an RFM-style or K-Means clustering analysis using these features to identify distinct customer segments for targeted marketing.

Report generated as part of Task 5 - Exploratory Data Analysis, Data Analyst Internship at DataX Labs.